

Chapter 12, Magnet Rampdown to Zero Field

12. Magnet Rampdown to Zero Field



⚠ DANGER

POTENTIAL FATAL INJURY!

BEFORE BEGINNING AND WHILE PERFORMING THIS PROCEDURE:

- MAKE SURE TO REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS.
- FULLY COMPLY WITH ALL REQUIRED ITEMS IN CHAPTER 1, SAFETY.
- HAVE ALL MR SAFETY TRAINED PERSONNEL COMPLY WITH THE SECOND PERSON SERVICE REQUIREMENTS AS SPECIFIED IN THE MR SERVICE SAFETY MANUAL, 5452735.



⚠ DANGER

POTENTIAL ASPHYXIATION HAZARD

PROCEDURE GENERATES ODORLESS, COLORLESS HELIUM GAS THAT DISPLACES OXYGEN IN THE AIR. MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS ON AND THE MAGNET ROOM DOOR IS PROPPED OPEN (OR THE HATCH IN A MOBILE VAN IS OPENED), BEFORE BEGINNING THIS PROCEDURE.



⚠ WARNING

EXPLOSION HAZARD!

LIQUID CRYOGENS CAN LIQUEFY ATMOSPHERIC OXYGEN, PRODUCING A HIGHLY COMBUSTIBLE OXYGEN LIQUID. NO SMOKING IN MAGNET AND CRYOGEN STORAGE ROOMS!



⚠ WARNING

ELECTRICAL SHOCK HAZARD!

DO NOT ALLOW EXPOSED BODY AREAS TO COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING WHILE INSERTING OR EXTRACTING THE MAIN LEAD EXTENSIONS.



⚠ WARNING

ELECTRICAL SHOCK HAZARD!

TOUCHING BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOWING THEM TO CONTACT EACH OTHER WHILE THE MAGNET IS AT FIELD WILL RESULT IN A RAPID DISCHARGE THROUGH THEIR CONTACT POINTS IF THE SWITCH HEATER IS ACTIVATED OR CIRCUIT RESISTANCE DEVELOPS AT THAT MOMENT. DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME. DO NOT ALLOW THE MAIN LEAD EXTENSIONS TO CONTACT EACH OTHER.



⚠ WARNING

ELECTRICAL SHOCK HAZARD!

CONTACT WITH CONNECTORS LEADING AN ENERGIZED POWER SUPPLY TO CAN CAUSE ELECTRICAL SHOCK.

- ALWAYS WEAR APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT (PPE) - ELECTRICALLY INSULATED GLOVES, NONFERROUS SAFETY SHOES AND SAFETY GLASSES/GOGGLES - WHILE PERFORMING THE CONNECTIONS FOR RAMPING PROCEDURE.
- FOLLOW LOCKOUT/TAGOUT PROCEDURES WHILE MAKING ELECTRICAL CONNECTIONS TO A MAIN POWER SUPPLY.

**! WARNING**

POTENTIAL COLD BURN HAZARD!

EXPOSURE TO CRYOGENS AND CONTACT WITH COLD OBJECTS IS POSSIBLE DURING THIS PROCEDURE. WEAR PROTECTIVE CLOTHING, NONABSORBENT GLOVES AND GOGGLES WHILE PERFORMING THIS PROCEDURE.

**! WARNING**

POTENTIAL PROJECTILE HAZARD!

FERROMAGNETIC OBJECTS IN A STRONG MAGNETIC FIELD CAN BECOME DANGEROUS PROJECTILES.

- REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. MAKE SURE ALL EQUIPMENT AND TOOLS USED IN THE MAGNET ROOM ARE NON-FERROMAGNETIC.
- LOCATE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND ROOM GEOMETRY ALLOWS.
- DO NOT ALLOW ANYONE CARRYING FERROMAGNETIC MATERIALS (INCLUDING CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES) INSIDE THE 5 GAUSS ZONE.

**! WARNING**

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH, RAPIDLY RELEASING CRYOGENIC GASES IF THE LIQUID HELIUM LEVEL DROPS TOO LOW DURING MAGNET RAMPDOWN. MAKE SURE THE HM MAGNET IS $\geq 75\%$, RD MAGNET IS $\geq 70\%$ AND UA MAGNET IS $\geq 98\%$ FULL OF LIQUID HELIUM BEFORE STARTING TO RAMP DOWN THE MAGNET.

**! WARNING**

POTENTIAL MAGNET QUENCH!

TURNING ON THE MAIN HEATER SWITCH WHILE PERFORMING RESISTANCE CHECKS WILL CAUSE A MAGNET QUENCH. MAKE SURE THE MAIN HEATER SWITCH STAYS OFF DURING RESISTANCE CHECKS.

**! WARNING**

POTENTIAL MAGNET QUENCH!

IF THE AXIAL SHIM HEATER SWITCH IS NOT "ON" DURING THE ENTIRE RAMPDOWN PROCESS, A MAGNET QUENCH CAN OCCUR AND IRREPARABLE SHIM COIL DAMAGE CAN RESULT. THE POWER SUPPLY WILL NOT PASS CURRENT TO THE MAIN LEAD EXTENSIONS WITH THE AXIAL SHIM HEATER OFF. MAKE SURE THE AXIAL SHIM HEATER SWITCH REMAINS "ON" DURING THIS ENTIRE PROCEDURE.

**! WARNING**

POTENTIAL MAGNET QUENCH!

INCORRECT CONNECTION POLARITY AND POWER SUPPLY CURRENT SETTINGS CAN CAUSE THE MAGNET'S MAIN COILS TO QUENCH AND CAN BURN UP THE POWER SUPPLY WHEN TURNING ON THE MAIN SWITCH.

**! WARNING**

POTENTIAL MAGNET QUENCH!

IMPROPER ENGAGEMENT/DISENGAGEMENT OF THE MAIN LEAD EXTENSIONS WITH THE MAIN LEAD PINS CAN CAUSE A RAMPED MAGNET TO QUENCH. INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND, IF THE MAGNET IS AT FIELD, ALLOW THEM TO COOL SUFFICIENTLY (A FOG OR WATER VAPOR FORMS AROUND THE LEAD EXTENSIONS) BEFORE THEY CONTACT THE MAIN LEAD PINS.

**CAUTION**

POTENTIAL MAGNET QUENCH!

UNACCEPTABLE INTERNAL CONTACT RESISTANCE AT THE MAIN LEAD EXTENSIONS CAN RESULT IN MAGNET QUENCH. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET, INCREASING BOIL-OFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING. MAKE SURE THE VOLTAGE READINGS ARE $\leq 100\text{mV}$ AT 750 AMPS FOR THE 1.5T MR450w (HM SERIES) MAGNET OR $\leq 100\text{mV}$ AT 750 AMPS FOR THE 1.5T LCCw (RD SERIES) MAGNET OR $\leq 40\text{mV}$ AT 600 AMPS FOR THE 3.0T MR750w (UA SERIES) MAGNET.

**CAUTION**

POTENTIAL MAGNET QUENCH!

THE MAGNET MAY QUENCH IF THE MAIN POWER SUPPLY EXPERIENCES LARGE OUTPUT FLUCTUATIONS AND/OR EXCESSIVE RIPPLE. MAKE SURE THE MAIN POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.

12.1. Magnet Rampdown to Zero Field, Tools and Test Equipment

Tools & Test Equipment

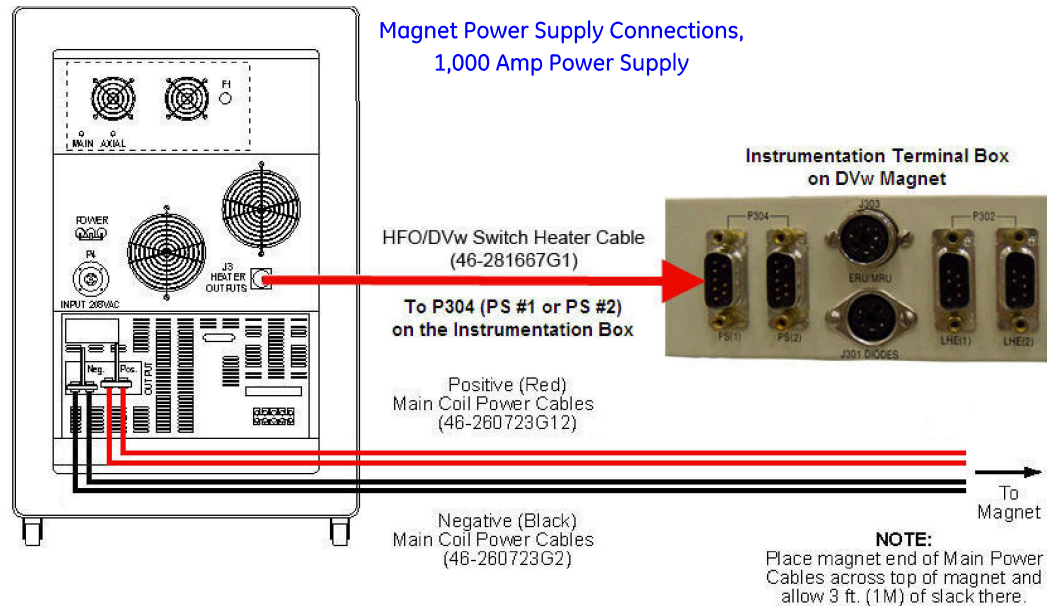
Part Number	Quantity	Description
5393536	1 per Person	Oxygen Sensor, O2 low alarm 19.5 percent volume, high alarm 23 percent volume
46-271137G1	1 per person	Cryogen Safety Wear Kit
NA	1 per person	Set: Nonabsorbent Protective Clothing (long sleeve shirt and pants)
NA	1 per person	Pair: Nonferrous Safety Shoes
NA	1 per person	Safety Glasses
NA	1 per person	Electrically Insulating Gloves
46-194427P320	1	Brass Master Padlock with Brass Shackle
2387081	1	Brass Master Transition Padlock
46-194427P322	1 pkg of 25	Black & White on Red Lockout Tag
46-194427P313	1	Transition LOTO Tag
46-194427P321	1	Line Cord Plug Cover, Plugs: ≤ 3 in. (76 mm) wide x ≤ 5.875 in. (149 mm) long, Cords: ≤ 1.25 in. (31 mm) diameter
5395802 or 5395802-100 or 46-260776G4	1	Auto Ramping Shim Power Supply (5395802 or 5395802-100) or 1000 Amp ESS 7.5-1000-2-D-1236 Main Power Supply, (46-260776G4)
46-294204G1 or 2287029	1	Fixed Main Lead Extension (46-294204G1) or Flexible Main Lead Extension (2287029, for low ceiling height sites)
5323611	1	DVw Ramp Lead Hold-Down Tool (included in Ramp Equipment Kit 46-260703G7)
2353394-3	1	1000 Amp Ramp Cable Kit with Ground
5395581	1	Y-Cable Extension (included in 1000 Amp Ramp Cable Kit 2353394-3)
NA	1	Digital Voltmeter (DVM), Volt- Ohm Meter (VOM) or non-contact voltage tester
46-251865G5	1	MetroLab Teslameter Kit
2217315	1	Auxiliary Rampdown Cable
2135558	1	Field Shim Kit
5113258 or 5112581	1 Kit	Titanium Tool Kit, Basic Set 5113258 or Titanium Tool Kit, Installation Set 5112581 or other non-ferromagnetic tools (magnets with magnetic fields $>1.5T$)
46-320273G4	1 Kit	Nonmagnetic Tool Kit, Inch/ Metric (magnets with magnetic fields $\leq 1.5T$ only) NOTE: Do not use these tools in the Magnet Room of magnets with magnetic fields $>1.5T$ whether or not the magnet is ramped.
Refer to Service Platform Set-Up	1	Service Platform with Side Shield/Side Protection Plate
NA	1	Service Methods documentation for the magnet/enclosure (available through the support documentation library at gehealthcare.com or your local GE Healthcare Service Representative)
NA	1	Cardboard Box or Nonferromagnetic Frame

12.2. Magnet Rampdown to Zero Field, Required Conditions

- 12.2.1. Make sure the area is secure and that all required Warning Signs are posted to meet the safety requirements stated in the Safety Chapter before starting rampdown procedure.
- 12.2.2. Removal of some magnet enclosure components may be required to complete the Connections for Ramping procedure. Refer to the appropriate Service Methods documentation (available through the support documentation library at gehealthcare.com or your local GE Healthcare Service Representative) before continuing with this procedure if cover removal is required.
- 12.2.3. Locate the Main Power Supply outside the magnet room, and remove all loose ferromagnetic materials from the Magnet Room before starting magnet rampdown.
- 12.2.4. Make sure the magnet room exhaust fan is on and operational before starting this procedure.
- 12.2.5. Make sure the magnet room door is propped open (or the hatch is opened if a mobile van) before starting this procedure.
- 12.2.6. Fill the magnet to the required helium level in conformance with Liquid Helium Fill before continuing with this Magnet Rampdown to Zero Field procedure.
 - ◆ 1.5T MR450w (HM Series): $\geq 75\%$
 - ◆ 1.5T LCCw (RD Series): $\geq 70\%$
 - ◆ 3.0T MR750w (UA Series): $\geq 98\%$
- 12.2.7. Set up MetroLab Teslameter (46-255839P2) and appropriate Teslameter Probe for single-point magnetic field strength monitoring in conformance with Preparations for Field Measurement.
 - ◆ 1.5T MR450w (HM Series): Use No. 5 Teslameter Probe (46-255839P104)
 - ◆ 1.5T LCCw (RD Series): Use No. 5 Teslameter Probe (46-255839P104)
 - ◆ 3.0T MR750w (UA Series): Use No. 6 Teslameter Probe (2295525-5)
- 12.2.8. Verify the Main Heater currents are set
 - ◆ 1.5T MR450w (HM Series): 600 to 620 mA
 - ◆ 1.5T LCCw (RD Series): 790 to 810 mA
 - ◆ 3.0T MR750w (UA Series): 660 to 720 mA
- 12.2.9. Disconnect and LOTO input power to the Main Power Supply and the Shim Power Supply. Verify that no voltage is present by using a DVM or equivalent measuring device.
- 12.2.10. Replace the Main Lead Contact Bands in conformance to Main Lead Extension Contact Band Replacement.
- 12.2.11. Connect the Switch Heater Cable (46-281667G1) between the Instrumentation Box (P304) at the magnet and the Main Power Supply (J3).

12.3. Magnet Rampdown with the Manual Ramping Power Supply (P/N 46-260776G4)

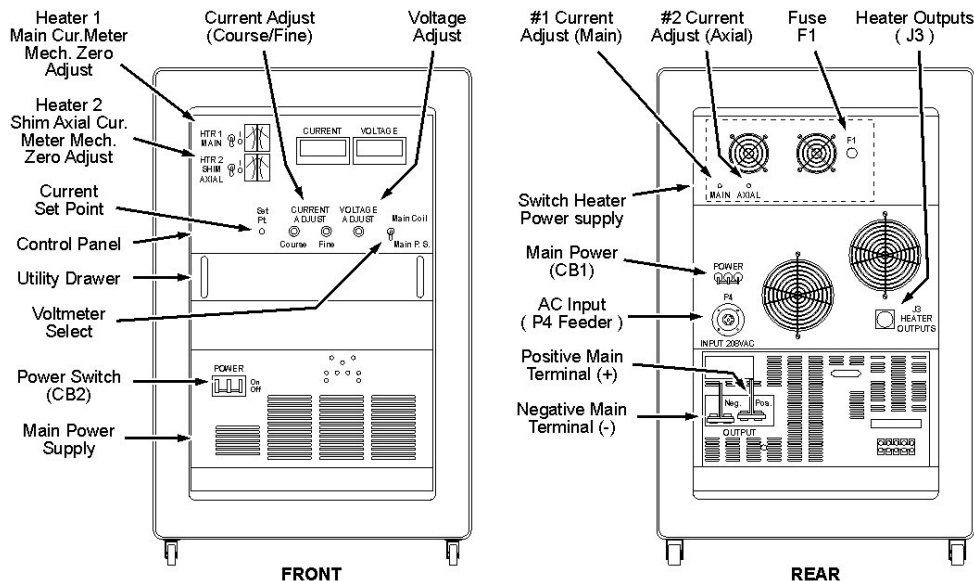
- 12.3.1. If you are using a manually ramping coil power supply (P/N 46-260776G4), connect electrical cabling to the Main Coil Power Supply as shown below.



- 12.3.2. Connect Power Supply Jumper Cable (as detailed in Appendix, "Connect Power Supply Jumper Cable"). Always install the Pre-Ramp Grounding Circuit and verify Main Lead Extension grounding to Cryostat ground before inserting or extracting any Main Lead Extension.
- 12.3.3. Make sure the input power cable for the Magnet Power Supply is disconnected and LOTO applied. Verify that no voltage is present by using a DVM or equivalent measuring device.

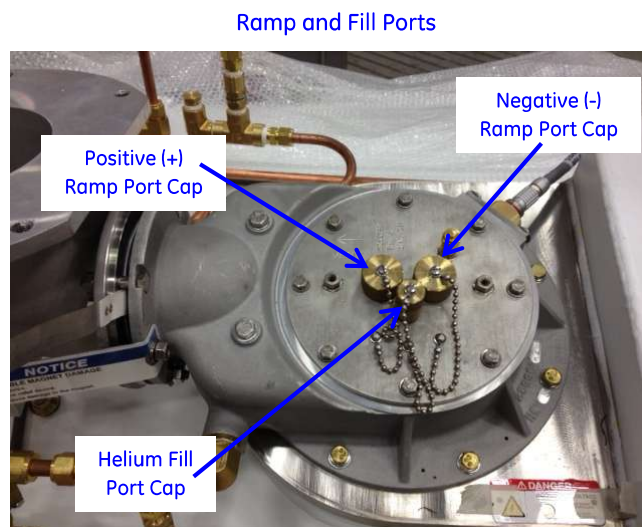
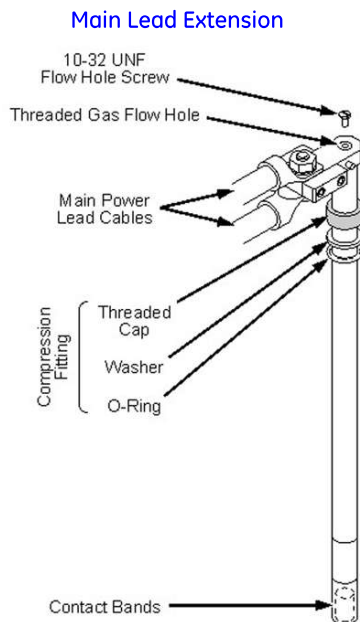
- 12.3.4. Connect the Main Power Supply and Main Lead Extensions to the magnet by making all cable connections in conformance with Connections for Ramping.
- 12.3.5. Set the Heater 1 Main and Heater 2 Shim Axial switches to the O (Off) position. (Shown below.)

ESS7.5-1000-2-D-1236 Superconducting Main Coil Service Power Supply Cabinet,
Manual Power Supply (46-260776G4)



- 12.3.6. Set the Current Adjust and Voltage Adjust controls to zero (fully CCW).
- 12.3.7. Remove LOTO and connect the input power to the Main Power Supply.
- 12.3.8. Disconnect Power Supply Jumper Cable. Once the Main Lead Extensions are inserted, disconnect and remove the Power Supply Jumper Cable from the power supply while wearing non-absorbent leather gloves with NO holes or tears. This removes the short across the power supply and enables magnet ramping up and down. The Power Supply Grounding Cable should remain connected while ramping up and down.
- 12.3.9. Set the Main Power and Power On switches to ON.
- 12.3.10. Connect a DVM or equivalent measuring device to the end of the Voltage Sense Leads.
- 12.3.11. Set the Current Adjust Controls on the Magnet Power Supply to maximum (fully CW).
- 12.3.12. Observe the Digital Current Meter and slowly turn the Voltage Control (CW) to the current value noted below through the Main Leads, Main Lead Extensions and the persistent Main Switch.
 - ◆ 1.5T MR450w (HM Series): 750 Amps
 - ◆ 1.5T LCCw (RD Series): 750 Amps
 - ◆ 3.0T MR750w (UA Series): 600 Amps
- 12.3.13. Obtain the voltage reading on the DVM. Adjust contact resistance (as detailed in Appendix, "Adjusting Contact Resistance") to achieve the required level. Higher resistances will add more heat to the magnet, increasing boil-off and possibly causing a quench during ramping. Do not proceed until after the DVM voltage is
 - ◆ 1.5T MR450w (HM Series) : ≤ 100 millivolts at 750 Amps
 - ◆ 1.5T LCCw (RD Series) : ≤ 100 millivolts at 750 Amps
 - ◆ 3.0T MR750w (UA Series): ≤ 40 millivolts at 600 Amps.

- 12.3.14. Set the power supply's Voltmeter Select Switch to the MAIN POWER SUPPLY position. This will display the output at the output lugs of the power supply monitored. A voltage of
 - ◆ 1.5T MR450w (HM Series): < 4.0V at 750A
 - ◆ 1.5T LCCw (RD Series) : < 4.0V at 750A
 - ◆ 3.0T MR750w (UA Series): < 3.5V at 600Aindicates acceptable system resistance.
- 12.3.15. If the voltage exceeds the above noted requirements during the test, follow the procedures above for decreasing contact resistance.
- 12.3.16. Turn the Current Adjust and Voltage controls OFF (fully CCW).
- 12.3.17. If a Quench occurs during change of magnetic field, immediately turn the Voltage Adjust and Current Adjust controls to zero.
- 12.3.18. Ice will build up around the Ramp Lead Hold-Down Tool flow holes during the rampdown procedure. Remove ice as needed to maintain helium gas flow through the flow holes.
- 12.3.19. Retrieve the latest final parking current.
- 12.3.20. Make sure the Axial Shim Heater (Ramp Power Supply) is on and the Main Leads are connected with the polarity indicated as shown below. Use DVM readings with the Voltage Sense Leads connected to the Main Lead Extensions for the steps below. This voltage approximates the Main Coil voltage, and will prevent the rampdown voltage from exceeding the recommended value.
- 12.3.21. Make sure the DVM is connected to the Voltage Sense Leads (shown below) from the Main Lead Extensions.



- 12.3.22. Set the Current Adjust Controls on power supply to 10 amps more than the parking current retrieved earlier.
- 12.3.23. Set the Voltage Control to adjust the power supply output current to the parking current value retrieved earlier.
- 12.3.24. Set the Heater 1 Main Switch to I (On).
- 12.3.25. Allow approximately one minute for the Main Switch to go normal.
- 12.3.26. When the Main Switch is normal, slowly decrease the Voltage Control (CCW) until -0.1 V is observed on the DVM.
- 12.3.27. Continue to adjust the power supply voltage, using the Voltage Control during rampdown in conformance with the voltages read on the DVM and Main Coil current ranges in the appropriate table below.

1.5T MR450w (HM Series) Rampdown Profile

Main Coil Voltage	Main Coil Current Range
-0.10 Volts	720 to 700 Amps
-0.30 Volts	700 to 675 Amps
-0.40 Volts	675 to 650 Amps
-0.50 Volts	650 to 625 Amps
-0.80 Volts	625 to 500 Amps
Gradually turn the Voltage Control to zero (fully CCW) then turn the Current Controls to zero (fully CCW).	500 to 0 Amps

1.5T LCCw (RD Series) Rampdown Profile

Main Coil Voltage	Main Coil Current Range
-0.10 Volts	Start to 720 Amps
-0.30 Volts	720 to 700 Amps
-0.40 Volts	700 to 675 Amps
-0.50 Volts	675 to 650 Amps
-0.80 Volts	650 to 625 Amps
-1.40 Volts	625 to 500 Amps
Gradually turn the Voltage Control to zero (fully CCW) then turn the Current Controls to zero (fully CCW).	500 to 0 Amps

3.0T MR750w (UA Series) Rampdown Profile

Main Coil Voltage	Main Coil Current Range
-0.25 Volts	> 580 Amps
-1.50 Volts	580 to 555 Amps
-3.00 Volts	555 to 480 Amps
-5.00 Volts	480 to 0 Amps
Gradually turn the Voltage Control to zero (fully CCW) then turn the Current Controls to zero (fully CCW).	At 0 Amps

- 12.3.28. Observe the voltmeter reading with the power supply's Voltmeter Select Switch in the MAIN COIL position. (A zero reading on the power supply's digital Voltage and Current Meters indicates that the magnet is fully discharged.)

- 12.3.29. When the magnet is fully discharged,
 - a. Set the Heater 1 Main Switch to O (Off).
 - b. Set the Main Power and Power On switches to OFF.
- 12.3.30. Disconnect and LOTO input power to the Main Power Supply. Verify that no voltage is present by using a DVM or equivalent measuring device. The Pressure Controller will maintain helium vessel pressure.
- 12.3.31. Connect Power Supply Jumper Cable (as detailed in Appendix, "Connect Power Supply Jumper Cable"). Always install the Pre-Ramp Grounding Circuit and verify Main Lead Extension grounding to Cryostat ground before inserting or extracting any Main Lead Extension.
- 12.3.32. After ramping down the magnet and disconnecting the main power supply, remove the Ramp Lead Hold-Down Tool.
- 12.3.33. Remove one Main Extension at a time in the following sequence:
 - a. Open Valve V2, and bleed vessel pressure down below 1.0 psig. (Do not allow pressure to go below 0.2 psig as read on the magnet pressure gauge or Magnet Monitor.) Close V2.
 - b. Remove all ice around the ramp lead port compression fitting on the first Main Lead Extension to be removed.
 - c. Unscrew the ramp lead port compression fitting, and remove the first Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
 - d. Remove all ice around the ramp lead port compression fitting on the other Main Lead Extension.
 - e. Unscrew the ramp lead port compression fitting, and remove the second Main Lead Extension from the magnet. Immediately replace the cap onto the Ramp Lead Port.
 - f. Check the Main Leads and Fill Port Cap for leaks.
 - g. Apply tape to cap chains to prevent spike noise (white pixels).
- 12.3.34. Disconnect Power Supply Jumper Cable. Once the Main Lead Extensions are inserted, disconnect and remove the Power Supply Jumper Cable from the power supply while wearing non-absorbent leather gloves with NO holes or tears. This removes the short across the power supply and enables magnet ramping up and down. The Power Supply Grounding Cable should remain connected while ramping up and down.